

# INEQUALITIES

Two real numbers or two algebraic expressions related by the symbols  $>$ ,  $<$ ,  $\leq$  or  $\geq$  form an inequality or inequation.

In other words, a statement involving the symbols  $>$ ,  $<$ ,  $\leq$  or  $\geq$  is called an inequality or inequation.

|        |               |                          |
|--------|---------------|--------------------------|
| $<$    | $\rightarrow$ | Less than                |
| $>$    | $\rightarrow$ | Greater than             |
| $\leq$ | $\rightarrow$ | Less than or equal to    |
| $\geq$ | $\rightarrow$ | Greater than or equal to |

## TYPES OF INEQUALITIES

A). **NUMERICAL INEQUALITY** : An inequality which does not involve any variable is called a numerical inequality.

e.g.  $5 > 3$  &  $8 < 28$

B). **LITERAL INEQUALITY** : An inequality which has variables is called a literal inequality.

e.g.  $a < 5$ ,  $x \geq 9$ ,  $y + z \geq 10$

C). **STRICT INEQUALITY** : An inequality which has only  $<$  or  $>$  is called a strict inequality.

e.g.  $5x + z < 0$

D). **SLACK INEQUALITY** : An inequality which has only  $\leq$  or  $\geq$  is called a slack inequality.

e.g.  $5x + z \leq 0$

## LINEAR INEQUALITY

An inequality is said to be linear, if the variable(s) occurs in first degree only & there is no term involving the product of the variables.

e.g.  $ax+b \geq 0$  ,  $ax+by+c > 0$  ,  $bx \leq 4$

LINEAR INEQUALITY IN ONE VARIABLE : A linear inequality which has only one variable, is called linear inequality in one variable.

e.g.  $ay+c > 0$  ,  $a \neq 0$

NOTE : An inequality in one variable, in which degree of variable is 2, is called quadratic inequality in one variable.

e.g.  $ax^2+bx+c \leq 0$  ,  $a \neq 0$

LINEAR INEQUALITY IN TWO VARIABLES : A linear inequality which has only two variables, is called linear inequality in two variables.

e.g.  $3x+10y \leq 0$  &  $4a+7b \geq 0$

### INTERVALS ON A NUMBER LINE

CLOSED INTERVAL : If  $a$  &  $b$  are real numbers, such that  $a < b$  then the set of all real numbers  $x$ , such that  $a \leq x \leq b$ , is called a closed interval & is denoted by  $[a, b]$ .

$$[a, b] = \{x : a \leq x \leq b, x \in \mathbb{R}\}$$

On the number line may be represented as



Here,  $a$  &  $b$  are included in the interval.

OPEN INTERVAL : If  $a$  &  $b$  are real numbers, such that  $a < b$  then the set of all real numbers  $x$ , such that  $a < x < b$ , is called an open interval & is denoted by  $(a, b)$ .

$$(a, b) = \{x : a < x < b, x \in \mathbb{R}\}$$

On the number line may be represented as



## SOLUTION OF AN INEQUALITY

- A). ADDITION OR SUBTRACTION : Some numbers maybe added or subtracted to/from both sides of an inequality.  
i.e if  $a > b$ , then for any number  $c$

$$a + c > b + c \quad \text{or} \quad a - c > b - c$$

- B). MULTIPLICATION OR DIVISION : If both sides of an inequality are multiplied (or divided) by the same positive numbers, then the sign of inequality remains same. But when both sides are multiplied (or divided) by the same negative numbers, then sign of inequality is reversed.

Let  $a, b$  &  $c$  be three real numbers, such that  $a > b$  &  $c \neq 0$ .

(i). If  $c > 0$ , then  $\frac{a}{c} > \frac{b}{c}$  &  $ac > bc$ .

(ii). If  $a > b$  &  $c < 0$ , then  $\frac{a}{c} < \frac{b}{c}$  &  $ac < bc$ .

## SOLUTION SET

The set of all solutions of an inequality is called the solution set of the inequality.

ALGEBRAIC SOLUTION OF LINEAR INEQUALITIES IN ONE VARIABLE : Any solution of an linear inequality in one variable is a value of the variable which makes it true statement.

Eg :  $x = 2$  is the solution of the linear inequality  $5x + 8 > 0$ .

## METHOD TO SOLVE A LINEAR INEQUALITY IN ONE VARIABLE

STEP I : Collect all terms involving the variable  $x$  on one side

and constant terms on other side with the help of above rules & then reduce it in the form  $ax < b$  or  $ax \leq b$  or  $ax > b$  or  $ax \geq b$ .

STEP II : Divide this inequality by the coefficient of variable  $x$ .  
This gives the solution set.

### SYSTEM OF INEQUALITIES IN ONE VARIABLE & THEIR SOL

Two or more inequalities taken together comprise a system of inequalities & the solution of the system of inequalities are the solutions common to all the inequalities comprising the system  
eg.  $x=3$  is a solution  $x > 2$  &  $x < 4$ .

NOTE : The solution set of a system of linear inequation in one variable is defined as the intersection of the solution set of linear inequalities in the system.

### TYPE I : WHEN TWO SEPARATE LINEAR INEQUALITIES ARE GIVEN

If the given system of inequalities comprises of two separate linear inequalities, then to solve these we use the following working steps.

STEP I : Solve each inequality separately & obtain their solution sets.

STEP II : Find the intersection of the solution sets obtained in STEP I

### TYPE II : WHEN INEQUALITIES OF THE FORM

$$a \leq \frac{cx+d}{e} \leq b, \text{ WHERE } a, b, c, d \in \mathbb{R}$$

This type of inequalities will be formed by combining the inequalities  $a \leq \frac{cx+d}{e}$  &  $\frac{cx+d}{e} \leq b$

To solve such type of inequalities, make the middle term free from constants.

TYPE III : WHEN INEQUALITY OF THE FORM  $|ax+b| \geq k$ ,  
WHERE  $a, b, k$  ARE REAL NUMBERS AND  $a \neq 0, k > 0$

If the given inequality is any of the form  $|ax+b| < k$  or  $|ax+b| > k$  or  $|ax+b| \geq k$  or  $|ax+b| \leq k$ , where  $a \neq 0$  &  $k > 0$ , then we solve the inequality with the help of following results.

If  $a > 0$  is any real number, then

$$(i). |x| < a \Rightarrow -a < x < a \\ |x| \leq a \Rightarrow -a \leq x \leq a$$

$$(ii). |x| > a \Rightarrow x > a \text{ or } x < -a \\ |x| \geq a \Rightarrow x \geq a \text{ or } x \leq -a$$

EXPANSION OF MODULUS FUNCTION

$$|x-a| = \begin{cases} x-a, & \text{if } x \geq a \\ -(x-a), & \text{if } x < a \end{cases}, \text{ where } a \text{ is any real no.}$$

TYPE IV : SOLUTION OF AN INEQUALITY, WHICH CONTAINS  
THE TERM OF THE FORM  $|x \pm a|$  AND  $|ax \pm b|$

STEP I : Put each modulus term equal to zero & find the corresponding value of  $x$ . The value of  $x$  are called critical points.

eg: If the given inequality involves the terms

$|x-a|$  and  $|x-b|$  then put

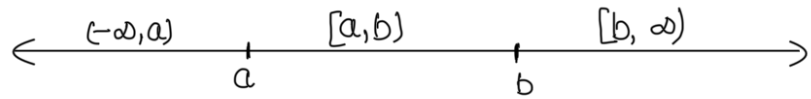
$$|x-a| = 0 \text{ and } |x-b| = 0$$

$$x-a = 0 \text{ and } x-b = 0$$

$$x=a \text{ and } x=b$$

STEP II : Divide the whole real line into disjoint intervals with the help of critical points.

eg: If  $x=a$  and  $x=b$  are critical points, such that  $a < b$ , then divide the real line as



STEP III<sup>rd</sup> : Consider the intervals obtained in Step II & solve the given inequality in each interval

STEP IV : Take the union of solution set .